

● MEDIUM

● ALGEBRA

● ANSWER KEY

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# Algebra

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08

**Q1****3**

The correct answer is 3. It's given that Cameron drove 60 miles per hour for  $x$  hours; therefore, the distance driven at this speed can be represented by  $60x$ . He then drove 50 miles per hour for  $y$  hours; therefore, the distance driven at this speed can be represented by  $50y$ . Since Cameron drove 210 total miles, the equation  $60x + 50y = 210$  represents this situation. If  $x = 1$ , substitution yields  $60(1) + 50y = 210$ , or  $60 + 50y = 210$ . Subtracting 60 from both sides of this equation yields  $50y = 150$ . Dividing both sides of this equation by 50 yields  $y = 3$ .

**Q2****294**

The correct answer is 294. Subtracting 18 from each side of the given equation yields  $\frac{6}{7}p = 36$ . Multiplying each side of this equation by  $\frac{7}{6}$  yields  $p = 42$ . Multiplying each side of this equation by 7 yields  $7p = 294$ . Therefore, the value of  $7p$  is 294.

**Q3****D**

D. 4

Choice D is correct. It's given that the function  $w$  models the volume of liquid, in milliliters, in a container  $t$  seconds after it begins draining from a hole at the bottom. The given function  $w(t) = 300 - 4t$  can be rewritten as  $w(t) = -4t + 300$ . Thus, for each increase of  $t$  by 1, the value of  $w(t)$  decreases by  $4(1)$ , or 4. Therefore, the predicted volume, in milliliters, draining from the container each second is 4 milliliters.

Choice A is incorrect. This is the amount of liquid, in milliliters, in the container before the liquid begins draining.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

**Q4****3**

The correct answer is 3. It's given that  $y = 9x + 12$ . Substituting  $9x + 12$  for  $y$  in the second equation in the system,  $x + 7y = 20$ , yields  $x + 7(9x + 12) = 20$ , which gives  $x + 63x + 84 = 20$ , or  $64x + 84 = 20$ . Subtracting 84 from each side of this equation yields  $64x = -64$ . Dividing each side of this equation by 64 yields  $x = -1$ . Substituting  $-1$  for  $x$  in the first equation in the system,  $y = 9x + 12$ , yields  $y = 9(-1) + 12$ , or  $y = 3$ . Therefore, the value of  $y$  is 3.

**Q5****C****C. 5**

Choice C is correct. Let  $a$  equal the number of 120-pound packages, and let  $b$  equal the number of 100-pound packages. It's given that the total weight of the packages can be at most 1,100 pounds: the inequality  $120a + 100b \leq 1,100$  represents this situation. It's also given that the helicopter must carry at least 10 packages: the inequality  $a + b \geq 10$  represents this situation. Values of  $a$  and  $b$  that satisfy these two inequalities represent the allowable numbers of 120-pound packages and 100-pound packages the helicopter can transport. To maximize the number of 120-pound packages,  $a$ , in the helicopter, the number of 100-pound packages,  $b$ , in the helicopter needs to be minimized. Expressing  $b$  in terms of  $a$  in the second inequality yields  $b \geq 10 - a$ , so the minimum value of  $b$  is equal to  $10 - a$ . Substituting  $10 - a$  for  $b$  in the first inequality results in  $120a + 100(10 - a) \leq 1,100$ . Using the distributive property to rewrite this inequality yields  $120a + 1,000 - 100a \leq 1,100$ , or  $20a + 1,000 \leq 1,100$ . Subtracting 1,000 from both sides of this inequality yields  $20a \leq 100$ . Dividing both sides of this inequality by 20 results in  $a \leq 5$ . This means that the maximum number of 120-pound packages that the helicopter can carry per trip is 5.

Choices A, B, and D are incorrect and may result from incorrectly creating or solving the system of inequalities.

**Q6****C**

C.  $P = \frac{10}{3}s - 2$

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Choice C is correct. The relationship between  $s$  and  $P$  can be modeled by a linear equation of the form  $P = ks + a$ , where  $k$  and  $a$  are constants. The table shows that  $P$  increases by 10 when  $s$  increases by 3, so  $k = \frac{10}{3}$ . To solve for  $a$ , substitute one of the given pairs of values for  $s$  and

$P$ : when  $s = 3$ ,  $P = 8$ , so  $8 = \frac{10}{3}(3) + a$ , which yields  $a = -2$ . The solution is therefore

$$P = \frac{10}{3}s - 2.$$

Choice A is incorrect. When  $s = 3$ ,  $P = 8$ , but  $3(3) + 10 = 19 \neq 8$ . Choice B is incorrect. This may result from using the first number given for  $P$  in the table as the constant term  $a$  in the linear equation  $P = ks + a$ , which is true only when  $s = 0$ . Choice D is incorrect and may result from using the reciprocal of the slope of the line.

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**Q7****B**

B. The average number of feet that the tree grew per year

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Choice B is correct. The height of the tree at a given time is equal to its height when it was planted plus the number of feet that the tree grew. In the given equation, 14 represents the height of the tree at the given time, and 6 represents the height of the tree when it was planted. It follows that  $2n$  represents the number of feet the tree grew from the time it was planted until the time it reached a height of 14 feet. Since  $n$  represents the number of years between the given time and the time the tree was planted, 2 must represent the average number of feet the tree grew each year.

Choice A is incorrect and may result from interpreting the coefficient 2 as doubling instead of as increasing by 2 each year. Choice C is incorrect. The height of the tree when it was 1 year old was  $2(1)+6=8$  feet, not 2 feet. Choice D is incorrect. No information is given to connect the growth of one particular tree to the growth of similar trees.

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**Q8****D**

D. 12,0

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Choice D is correct. The given function  $f$  is a linear function. Therefore, the graph of  $y = fx$  in the  $xy$ -plane has one  $x$ -intercept at the point  $k, 0$ , where  $k$  is a constant. Substituting 0 for  $fx$  and  $k$  for  $x$  in the given function yields  $0 = 7k - 84$ . Adding 84 to both sides of this equation yields  $84 = 7k$ . Dividing both sides of this equation by 7 yields  $12 = k$ . Therefore, the  $x$ -intercept of the graph of  $y = fx$  in the  $xy$ -plane is 12,0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

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**Q9****D**

D. 18

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Choice D is correct. Adding the second equation to the first equation in the given system of equations yields  $3y - 3y = 4x + 9x + 17 - 23$ , or  $0 = 13x - 6$ . Adding 6 to each side of this equation yields  $6 = 13x$ . Multiplying each side of this equation by 3 yields  $18 = 39x$ . Therefore, the value of  $39x$  is 18.

Choice A is incorrect. This is the value of  $-39x$ , not  $39x$ .

Choice B is incorrect. This is the value of  $-13x$ , not  $39x$ .

Choice C is incorrect. This is the value of  $13x$ , not  $39x$ .

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## Q10

D

D.  $a + b \leq 164$   
 $a \leq 3b$

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Choice D is correct. It's given that a city employee will plant azaleas and boxwoods in a park, where  $a$  is the number of azaleas that will be planted and  $b$  is the number of boxwoods that will be planted. It's also given that there will be no more than 164 total bushes planted, which can be represented by the inequality  $a + b \leq 164$ . It's also given that the number of azaleas planted will be at most three times the number of boxwoods planted, which can be represented by the inequality  $a \leq 3b$ . Therefore, the system of inequalities containing  $a + b \leq 164$  and  $a \leq 3b$  best represents this situation.

Choice A is incorrect. The inequality  $a + b \geq 164$  represents a situation where at least 164 total bushes will be planted, not that there will be no more than 164 total bushes planted. Also, the inequality  $3a \geq b$  represents a situation where the number of boxwoods that will be planted is at most three times the number of azaleas that will be planted, not that the number of azaleas planted will be at most three times the number of boxwoods planted.

Choice B is incorrect. The inequality  $a + b \geq 164$  represents a situation where at least 164 total bushes will be planted, not that there will be no more than 164 total bushes planted.

Choice C is incorrect. The inequality  $3a \geq b$  represents a situation where the number of boxwoods that will be planted is at most three times the number of azaleas that will be planted, not that the number of azaleas planted will be at most three times the number of boxwoods planted.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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